Review proposal for Foundations of Physics

From local quantum-field measurements to certified resources

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Proposed article type: Critical narrative review.

Central question. Which assumptions allow a local quantum-field measurement to justify a claim of resource extraction? The proposed review follows the inference from a field state and a causally admissible interaction to a reduced probe state, a resource criterion, and a calibrated record. It distinguishes prediction of a resource, certification of its presence, and attribution of its origin to the field.

Fit to the journal. The article connects the conceptual foundations of relativistic measurement with the operational interpretation of quantum resources. Its primary concerns are locality, state updating, subsystem assignment, and what evidence supports a physical claim. Communication and metrology are included as conditional tests of utility, rather than as separate application surveys.

Relation to existing reviews. The closest work is M. Papageorgiou and D. Fraser, *Eliminating the 'Impossible': Recent Progress on Local Measurement Theory for Quantum Field Theory*, Foundations of Physics **54**, 26 (2024), https://doi.org/10.1007/s10701-024-00756-8. That review organizes detector, algebraic, and histories-based measurement proposals around impossible-measurement scenarios. The present proposal builds on those distinctions and asks what further premises connect an allowed operation or a record to a resource claim. It does not claim to replace the earlier review or to introduce a universal resource theory.

Specific comparisons.

- Identical measurement effects can correspond to different local instruments and different resource-bearing outputs. Selective updating is distinguished from nonselective signalling.
- Data-processing bounds require a specified field algebra, product assumptions, and regions containing the full interaction supports. Shrinking the probe's spatial size does not shrink a fixed-duration interaction to an equal-time point.
- Fixed-scenario contextual fractions and preparation-family generalized noncontextuality concern different operational structures. Recent preprints are reported with their hypotheses and publication status.
- A raw residual negativity can be maximal on a biseparable mixed state. Exact local-channel invariance and nonuniform perturbative expressions support different physical interpretations.
- Composite observables, model truncation, calibration, and finite sampling impose separate conditions. A directly measured witness and a predicted perturbative witness require different evidence.

Organization. The main article has eight sections: (1) Introduction; (2) Local operations, records, and causal realization; (3) Field regions and accessible probe resources; (4) Entanglement harvesting and causal attribution; (5) What a resource criterion certifies; (6) The coupled observable and the calibrated record; (7) What certification implies for a task; (8) Conclusions.

Supporting material and scope. Seven supplementary sections provide the instrument example, finite-mode and compact-support benchmarks, contextuality transport argument, finite-

state resource diagnostics, separated composite kernel, and finite-sample witness proof. Reproducibility files state what was checked and distinguish exact algebra, perturbative diagnostics, and numerical convergence. The scope is the local measurement and certification problem; background-specific scattering calculations and detailed network engineering lie outside it. No new experiment or all-orders continuum harvesting calculation is claimed.

The manuscript and supplementary material have been prepared. I would welcome advice on whether this review is suitable for submission to Foundations of Physics and whether the proposed scope should be adjusted before formal submission.